

Everything You May Like To Know About Infrared **WATER FAUCETS**



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Infrared (IR) water faucets (touch-free faucets) look modernistic, and can significantly increase the style level in a luxurious bathroom. Nowadays, IR water faucets are widely used in hospitals and in other areas with industrial or public sanitary installations because traditional water faucets can collect bacteria and viruses, as these are constantly being touched by many different people. Further, IR water faucets are a proven solution to save water.

System overview

Although it looks like a sleek accessory in a stylish bathroom, there is a lot happening under the sink with an IR water faucet. In principle, an IR water faucet combines four key components: IR sensor module, solenoid valve, control box and power source.

IR sensor module. An IR sensor module is usually visible and looks like a small lens on the front of the faucet. The sensor module (covered by a plastic IR sensor shell) holds one transmitter component and one receiver component.

The transmitter (usually an IR LED) continuously emits invisible light to scan whether there is an object in front of the water faucet. If a hand, for instance, enters within the range, it reflects the transmitted IR light. The receiver (usually a photodiode or phototransistor) recognises this and opens the solenoid valve with the help of

associated controller electronics.

In short, if the sensor feels the presence of an object in front of the tap, it actuates the solenoid valve to initiate the flow of water. When the object is no longer present, it deactivates the solenoid valve to terminate the water flow, usually after a few seconds.

Solenoid valve. A solenoid valve physically starts and stops the water flow by transforming electrical energy into motion. A standard solenoid valve allows water flow when DC supply holds a plunger at its open position. In the absence of DC supply, the plunger returns to its initial (closed) position to prevent the flow of water.

Note that, pulse-type (latching-type) solenoid valves are also used in today's IR faucets. Here, the solenoid valve is initially pulsed to start the water flow (plunger is driven into the vicinity of a permanent magnet, which, in turn, holds the plunger in open position). In order to return the plunger to its closed position, the solenoid valve is once again pulsed, but this time with a reverse-polarity signal.

Control box. A control box is like a control hub that sits or is mounted underneath the sink as part of a typical IR water faucet installation process. The control box is a rigid, water-proof enclosure that accommodates almost all requisite components/parts (except IR sensor module) like solenoid valve, electronic control circuitry and power source. Most control hubs can be powered by built-in battery packs or by mains via external AC-DC power adaptors.

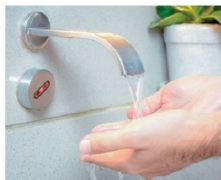


Fig. 1: IR water faucet



Fig. 2: IR sensor module



Fig. 3: Solenoid valve



Fig. 4: Control box

Overview breakdown

IR water faucets are created by combining the four key components. Although there are variations, key components, each with a distinct function, once com-